

Apex Inventory

UI Requirement Specification

Layout & Navigation Shell

Route: All routes (wrapper component)

Document Version: 1.0.0

Project: ReactExample (Apex Store Inventory Management)

1. Overview

The Layout component is the persistent application shell that wraps every page in Apex Inventory. It provides the sidebar navigation, brand identity, real-time alert indicators, and a main content area where individual page components are rendered via React Router's Outlet. Users never interact with Layout as a standalone page; it is the structural frame for the entire application.

2. Purpose

- Provide consistent navigation across all inventory management screens.
- Display global inventory health indicators (low stock count, pending orders).
- Maintain brand identity with the 'Apex Inventory' logo and version footer.
- Host child route content without re-mounting the sidebar on navigation.

3. UI Structure

3.1 Sidebar (Left Panel)

- Brand logo area: Store icon + 'Apex Inventory' text.
- Vertical navigation list with five links (see Navigation Items below).
- Low Stock Alert banner (conditional): shown at bottom of nav when any item has quantity ≤ 5 .
- Footer: displays 'Apex Store v1.0.0'.

3.2 Main Content Area (Right Panel)

- Renders the active page component through `<Outlet />`.

- Scrollable region for page-specific content (Dashboard, Inventory, etc.).

4. Navigation Items

Label	Route / Behavior
Dashboard	/ - Analytics overview, KPIs, charts
Inventory Items	/inventory - Search, filter, manage stock
Add Stock	/add-stock - Register incoming stock
Categories & Layout	/categories - Category CRUD and store map
Replenish Orders	/orders - Place and track supplier orders

5. Behavior & Interactions

5.1 Active Route Highlighting

Each NavLink uses React Router's `isActive` callback. The currently active route receives the 'active' CSS class, visually distinguishing it from inactive links. Dashboard uses the 'end' prop so only exact '/' matches are highlighted (not child paths).

5.2 Pending Orders Badge

On the 'Replenish Orders' nav item, a numeric badge appears when `pendingOrdersCount > 0`. The badge shows the count of orders with status 'pending'. Badge styling: secondary color background, dark text, rounded corners.

5.3 Low Stock Alert Banner

When `lowStockCount > 0`, a red-tinted alert box appears above the sidebar footer. It displays an `AlertTriangle` icon and text: 'Low Stock Alert' with '{count} items need attention'. Low stock is defined as items with quantity ≤ 5 (includes out-of-stock items at quantity 0).

5.4 Stats Refresh on Navigation

A `useEffect` hook re-fetches items and orders whenever `location.pathname` changes. This ensures sidebar badges and alerts stay current after the user performs actions on other pages (e.g., placing an order or adjusting stock).

6. Data Dependencies

API Call	Purpose
GET /api/items	Count items with quantity ≤ 5 for low stock alert
GET /api/orders	Count pending orders for nav badge

7. Error Handling

If API calls fail, errors are logged to console. The sidebar still renders; badge counts default to 0. No user-facing error message is shown in the Layout component.

8. Technical Notes

- Component file: `src/components/Layout.tsx`
- Icons: `lucide-react` (Store, LayoutDashboard, Package, PlusCircle, MapPin, Truck, AlertTriangle)
- Routing: nested under `<Route path="/" element={}<Layout />>` in `App.tsx`